Pierre **Courteille**

Cloud Architect · DevOps Engineer

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Education

University of Montpellier — *Montpellier, France*

Master's, Mathematics & Computer Science (Master MOCA — modeling, optimization, combinatorics, algorithms) — *2012–2015*

University of Nantes — *Nantes, France*

Bachelor's, Mathematics & Computer Science — *2009–2012*

Skills

DevOps & platforms	AWS (Organizations, CloudFormation, CDK, CloudWatch, Systems Manager), Docker, Kubernetes (EKS), ECS Fargate, GitLab CI/CD, CodePipeline, CodeBuild, Jenkins, Terraform
Cloud architecture	Well-Architected pillars, multi-account landing zones, microservices, event-driven design, API-first and serverless (Lambda, API Gateway), multi-AZ design, high availability, multi-region and disaster recovery
Networking & connectivity	VPC (subnets, route tables, NACLs, security groups), Transit Gateway (hub-and-spoke, VPC/VPN/Direct Connect attachments, inter-region peering), hybrid connectivity (Site-to-Site VPN, Direct Connect patterns), Route 53, VPC endpoints, PrivateLink
Security & governance	IAM, IAM Identity Center, SCPs, least privilege, KMS, WAF, Security Hub, AWS Config, VPC endpoints, secure cross-account access
Data & integration	RDS, DynamoDB, Redshift, ElastiCache, S3, messaging and integration (SQS, SNS, EventBridge), log/query (Athena, CloudWatch Logs Insights)
Observability & FinOps	CloudWatch OAM, Grafana, X-Ray, centralized logging/metrics/alerting, budgets, Cost Explorer, CUR-based reporting
Programming & automation	Python (incl. Boto3), C#, Node.js, JAVA, C/C++, .NET Core, shell scripting, infrastructure automation (CDK, Terraform, CloudFormation, AWS SAM)
Languages	French, English, Spanish

Certificates

- Jul 2025 **AWS Certified DevOps Engineer – Professional (DOP-C02)**, Amazon Web Services (AWS)
- Jul 2025 **AWS Certified Solutions Architect – Associate (SAA-C03)**, Amazon Web Services (AWS)
- Jul 2025 **AWS Certified Developer – Associate (DVA-C02)**, Amazon Web Services (AWS)
- Jun 2025 **AWS Certified Cloud Practitioner (CLF-C02)**, Amazon Web Services (AWS)

Experience

Qim info / International Electrotechnical Commission (IEC)

Geneva, Switzerland

CLOUD ENGINEER *Oct. 2025 - Present*

Design and operation of multi-account AWS environments for several applications, with emphasis on security, scalability, and automation. Partner with product and security stakeholders to keep tenant isolation, delivery speed, and operability aligned.

- **SaaS architecture (Silo model):** multi-tenant design (1 tenant = 1 AWS account) via AWS Organizations; strong isolation (dedicated VPC patterns, IAM boundaries, data perimeter) and reduced blast radius for regulated workloads.
- **Multi-account governance & security:** OU layout, SCP guardrails, centralized logging and audit expectations, least-privilege IAM roles, secure cross-account access (e.g. assume-role patterns, resource shares).
- **IaC & delivery:** AWS CDK platform to deploy Docker microservices on ECS Fargate; standardized application pipelines, reusable constructs, automated tenant onboarding (accounts, baseline stacks, networking hooks).
- **Observability (multi-account / multi-region):** CloudWatch OAM, dedicated monitoring account, Grafana; centralized metrics/logs, SNS alerting, synthetic checks for critical user journeys across regions.
- **Operations & reliability:** VPC lifecycle, WAF on CloudFront, EC2 → ECS migrations; log analytics with Athena and CloudWatch Logs Insights; troubleshooting runbooks and post-incident follow-up.
- **Cost & efficiency:** tagging and allocation basics, right-sizing discussions with owners, visibility into spend drivers (where FinOps tooling applies).

Stack: AWS, AWS Organizations, CDK, Terraform, ECS Fargate, Docker, CloudWatch, Grafana, WAF, Athena, Python.

Premaccess

Paris, France

CLOUD ENGINEER (FREELANCE) *Jun. 2019 - Sep. 2025*

Support for multiple clients on AWS cloud transformation: migration through optimization and scale-out, with strong automation, security, and performance focus. Work spanned greenfield builds, legacy modernization, and operational hardening.

- **Cloud migration & scalability:** led migrations and architecture evolution toward ECS, EKS, and serverless (Lambda, API Gateway); managed data on RDS, DynamoDB, Redshift, ElastiCache; capacity and scaling patterns per workload.
- **Network & security:** multi-AZ VPC (public / private / isolated subnets), Security Groups, NACLs, VPC Endpoints; IAM Identity Center, SCPs, KMS encryption; reduced exposure and clearer network segmentation.
- **CI/CD & IaC:** CodePipeline, CodeBuild, GitLab CI, Jenkins; CDK, CloudFormation, Terraform for repeatable environments and peer-reviewable change.
- **Observability & FinOps:** CloudWatch, X-Ray, Config, Security Hub; dashboards for budgets, CUR, Cost Explorer; helped teams connect technical choices to spend.
- **Automation & performance:** internal provisioning tooling (Python, Boto3); C# refactor (+40% throughput, 30% lower latency); AWS SAM for serverless; L2/L3 support, architecture reviews, and knowledge transfer.
- **Stakeholder work:** workshops with client teams, prioritization of security vs. velocity trade-offs, documentation of standards (naming, tagging, baseline modules).

Stack: AWS, Python, Node.js, .NET Core, Docker, Kubernetes, CDK, Terraform.

ISIKA

Paris, France

.NET INSTRUCTOR & SOFTWARE ENGINEER (FREELANCE) *Mar. 2020 - Apr. 2023*

C# and .NET training with a focus on OOP and application design. Combined formal teaching with hands-on engineering so learners could ship small end-to-end applications.

- **Technical training:** C# fundamentals, OOP (classes, inheritance, polymorphism), desktop development (WinForms), debugging and code structure.
- **Application development:** guided learners through full apps including data persistence (LINQ, Entity Framework, MySQL), layering, and basic testing mindset.
- **Pedagogy:** project follow-up, individual coaching, rubrics for assessed deliverables; adapted pace to professional audiences.
- **Course material:** maintained exercises and examples aligned with workplace scenarios (CRUD, validation, simple UI patterns).

Stack: C#, HTML5, Entity Framework, LINQ, MySQL, WinForms.

Bassetti

Kolkata area, India

SOFTWARE QUALITY ASSURANCE MANAGER *Feb. 2016 - May 2018*

Built and led a QA organization in India: team leadership, large-scale test automation, and DevOps practices. Focus on throughput, traceability, and predictable releases.

- **Hiring & leadership:** recruited, trained, and led 25+ QA engineers; career paths, skills matrix, and on-boarding for distributed teams.
- **Automation & tooling:** co-designed an internal platform executing 10,000+ API tests/hour; improved coverage, flakiness reduction, and reporting for leadership.
- **Quality & delivery:** tight collaboration with development; automated feedback loops, daily test runs, shorter regression cycles before releases.
- **Process & DevOps:** introduced DevOps-oriented practices (shared ownership of quality gates, CI integration), clearer QA–dev handoffs, faster cadence without sacrificing critical checks.
- **Metrics:** tracked defect trends, escape rate, and automation ROI to steer investment in the right layers of the stack.

Stack: C#, .NET, Selenium, Jenkins, APIs, XML, Excel.

Kronos Incorporated (Ad Opt)

Montreal, Canada

OPTIMIZATION ENGINEER — *Jan. 2015 - Jan. 2016*

Ad Opt (Kronos) — optimization for airline crew planning (cost and operational efficiency). Mixed R&D, heavy computational experiments, and customer-facing delivery.

- **Algorithmic optimization:** performance gains via column generation and shortest-path strategies on large instances; profiling and memory-aware implementations in C/C++.
- **Validation:** intensive testing on thousands of instances; stress cases for edge constraints and stability before production rollout.
- **Production:** customer deployments, integration support, and tuning with operations teams.
- **R&D:** presentations at GERAD; contributions to scientific publications; knowledge sharing on modeling choices and solver behavior.
- **Engineering practice:** Linux toolchain, GDB, Git workflows; collaboration with researchers and product on requirement formalization.

Stack: C, C++, Linux, GDB, Git.